Solution of Tutorial Sheet 7

(1) Determine zero's, singular points, nature of singular points & poles of the fn.

(a) $$f(z) = \frac{z^2}{(z^2+1)^2(z+3)}$$

$$= \frac{z^2}{(z+i)^2(z-i)^2(z+3)}$$

i.e. $z = i, -i, 3 \longrightarrow$ are singular points & they are isolated singularity.

In fact $i, -i$ are pole of order 2.

(Since $f(z) = \dfrac{\phi_1(z)}{(z+i)^2}$ where

$$\phi_1(z) = \frac{z^2}{(z-i)^2(z+3)}$$

analytic at $z=-i$ & $\phi_1(-i) \neq 0$

$f(z) = \dfrac{\phi_2(z)}{(z-i)^2}$, where

$$\phi_2(z) = \frac{z^2}{(z+i)^2(z+3)}$$

analytic at $z=i$ and

$\phi_2(i) \neq 0$.

$z=-3$ is a pole of order 1 (i.e. simple pole)

Since $f(z) = \dfrac{\phi_3(z)}{(z+3)}$ where

$$\phi_3(z) = \frac{z^2}{(z-i)^2(z+i)^2}$$

analytic at $z=-3$ & $\phi_3(-3) \neq 0$

$f(z) = (z-0)^2 g(z)$

$= \cancel{(z-0)}$ where $g(z) = \dfrac{1}{(z^2+1)^2(z+3)}$ is analytic $z_0 = 0$ & non zero at $z_0 = 0$.

i.e. $f(z)$ has zero of order 2 at $z=0$.

(ii) $f(z) = 1/z^2$

has a pole of order (2).

$z = 0$ is a isolated singularity.

NO zero

(iii) $f(z) = \dfrac{e^z}{(z - \sin z)}$

Now

$$z - \sin z = \cancel{z} - \left[\cancel{z} - \frac{z^3}{\lfloor 3} + \frac{z^5}{\lfloor 5} - \cdots\right]$$

$$= \frac{z^3}{\lfloor 3} - \frac{z^5}{\lfloor 5} + \frac{z^7}{\lfloor 7} - \cdots$$

$$= z^3 \underbrace{\left[\frac{1}{\lfloor 3} - \frac{z^2}{\lfloor 5} + \frac{z^5}{\lfloor 7} - \cdots\right]}_{g(z)}$$

$\boxed{z - \sin z = z^3 g(z)}$ where $g(z)$ is analytic at z_0 & $g(0) \neq 0$

Now $f(z) = \dfrac{e^z}{z^3 g(z)}$

$f(z) = \dfrac{e^z/g(z)}{(z-0)^3}$ i.e. has pole of order 3

$z = 0 \rightarrow$ isolated singular point. & $f(z)$ has no zero.

(iv) $f(z) = \dfrac{\sin z}{z^3}$

has a pole of order 3 at $z=0$.

$z=0$ is isolated singularity.

$z = n\pi$ when $n \in \mathbb{Z} \setminus \{0\}$ are ~~simple~~ zero's of order 1.

(v) $f(z) = \operatorname{Ln}(z^2-1)$

$= \operatorname{Ln} g(z)$, $\quad g(z) = z^2-1 = x^2-y^2+2ixy-1$

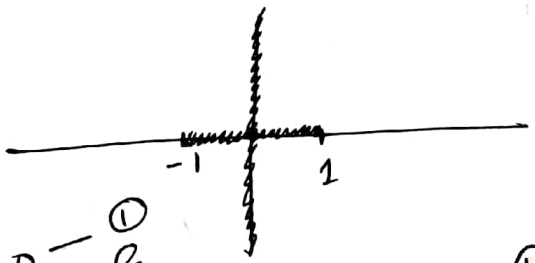
where $g(z) = z^2-1$

$= (z+1)(z-1)$

Branch cut $\Rightarrow$

$\operatorname{Re} g(z) \le 0$ &

$\operatorname{Im}(g(z)) = 0$



i.e. $x^2-y^2-1 \le 0$ — (I) & $2xy = 0$ — (II)

$x=0$ or $y=0$

If $x=0$ then by (I),

$-y^2-1 \le 0$

$y^2+1 \ge 0$

$y^2 \ge -1$

$\boxed{y^2+1 \ge 0}$, $x=0$

If $y=0$, then by (I) is

$x^2-1 \le 0$

$\boxed{x^2 \le 1}$, $y=0$

1 & -1 are non isolated singularity.

Non isolated singular points are $\Longrightarrow$ Y-axis.

on X-axis, it is $-1 \le X \le 1$

(vi) $f(z) = e^{1/z^2}$

$= 1 + \frac{1}{z^2} + \frac{1}{12 z^4} + \frac{1}{13} \frac{1}{z^6} + \cdots$

has essential singularity at $z = 0$

NO zero.

(vii) $f(z) = \sum_{n=0}^{\infty} n(-1)^n z^n$ (we have a taylor series expansion)

$= 0 - z + 2z^2 - 3z^3 + \cdots$

$f(z) = z g(z)$

where $g(z) = [-1 + 2z - 3z^2 + \cdots]$

has zero at $z = 0$ of order 1.

$(g(0) \neq 0)$

radius of convergent.

$$R = \frac{1}{\lim_{n \to \infty} \left| \frac{a_{n+1}}{a_n} \right|}$$

$$= \frac{1}{\lim_{n \to \infty} \left| \frac{(n+1)(-1)^{n+1}}{n(-1)^n} \right|}$$

$$= \frac{1}{\lim_{n \to \infty} \left| \frac{n+1}{n} \right|} = 1$$

i.e. $\boxed{|z - 0| < 1}$

$$\frac{1}{(1+z)} = 1 - z + z^2 - z^3 \cdots \qquad |z|<1$$

$$\frac{-1}{(1+z)^2} = 0 - 1 + 2z - 3z^2 + 4z^3 - \cdots \qquad |z|<1$$

$$\Rightarrow \frac{-z}{(1+z)^2} = -z + 2z^2 - 3z^3 + 4z^4 - \cdots \qquad |z|<1$$

$$\cancel{\frac{z}{(1+z)^2} = 1 - 2z + 3z^2 - 4z^3 + \cdots} \qquad |z|<1$$

$$\Rightarrow \boxed{\frac{-z}{(1+z)^2} = \sum_{n=0}^{\infty} n(-1)^n z^n} \qquad |z|<1$$

has a pole of order 2 at $z=-1$

Laurent series exp about $z=-1$

$$\frac{-z}{(1+z)^2} = \frac{-(z+1)+1}{(1+z)^2}$$

$$= -\frac{1}{(1+z)} + \frac{1}{(1+z)^2}$$ (i.e. pole of order 2)

(2) find out residue of $f(z)$ at z_0 $\longrightarrow$

(a) $f(z) = \dfrac{z+1}{z^2+9}$ at $z = 3i, -3i$

(b) $f(z) = \dfrac{z^3+2z}{(z-i)^3}$ at $z=i$

(c) $f(z) = z \sin 1/z$ $\quad \underline{\underline{z_0 = 0}}$

$$= z\left[1/z - \frac{1}{z^3 \lfloor 3} + \frac{1}{z^5 \lfloor 5} - \cdots\right]$$

$$= 1 - \frac{1}{z^2 \lfloor 3} + \frac{1}{z^4 \lfloor 5} - \cdots \qquad \underline{\underline{0<|z|<\infty}}$$

$$\operatorname*{Res}_{z=z_0} f(z) = \operatorname*{Res}_{z=0} f(z) = \underline{\underline{b_1 = 0}} \ \#$$

(d) $f(z) = \cot z$, $\underline{\underline{z_0 = 0}}$

$f(z) = \dfrac{\cos z}{\sin z}$ has simple pole at $\underline{\underline{z = n\pi \ \forall n \in \mathbb{Z}}}$.

$$\operatorname*{Res}_{z=0} f(z) = \lim_{z\to 0} (z-0)\cdot\frac{\cos z}{\sin z} = \lim_{z\to 0} \frac{\cos z}{(\sin z/z)} = 1 \ \#$$

OR

$f(z) = \dfrac{p(z)}{q(z)} = \dfrac{\cos z}{\sin z}$, $p(z)$ & $q(z)$ both are analytic $\underline{\underline{\text{at } z=0}}$. $p(0) = \cos 0 = 1 \neq 0$

$q(0) = \sin 0 = 0$, $q'(0) \neq 0$.

i.e. 0 is a simple pole of $\dfrac{p(z)}{q(z)}$ & $\operatorname*{Res}_{z=0} \dfrac{p(z)}{q(z)} = \dfrac{p(0)}{q'(0)} = 1 \ \#$

(e) $f(z) = z e^{\pi/z}$ at $z=0$

$$= z\left[1 + \frac{\pi}{z} + \frac{\pi^2}{z^2 \lfloor 2} + \frac{\pi^3}{z^3 \lfloor 3} + \cdots\right]$$

$$= z + \pi + \frac{\pi^2}{z \lfloor 2} + \frac{\pi^3}{z^2 \lfloor 3} + \cdots \qquad \underline{0<|z|<\infty}$$

i.e. $\operatorname*{Res}_{z=0} f(z) = b_1 = \dfrac{\pi^2}{\lfloor 2} = \dfrac{\pi^2}{2} \ \#$

Note:-

(i) Residue at a pole of order k

$$\boxed{\operatorname*{Res}_{z=z_0} f(z) = \frac{1}{\lfloor k-1} \lim_{z\to z_0} \frac{d^{k-1}}{dz^{k-1}} (z-z_0)^k f(z)}$$

OR

$$= \frac{1}{\lfloor k-1} \lim_{z\to z_0} g^{(k-1)}(z)$$

$$\boxed{\text{where } g^{k-1}(z) = \frac{d^{k-1}}{dz^{k-1}} (z-z_0)^k f(z)}$$

(ii) Residue at a pole of order 1 (i.e. simple pole)

$$\boxed{\operatorname*{Res}_{z=z_0} f(z) = \lim_{z\to z_0} (z-z_0) f(z)}$$

(3) Compute the integral by Residue theorem →

(a) $\displaystyle\int_{|z|=1} z^2 \sin 1/z \, dz$

$= 2\pi i \, (b_1) \rightarrow \operatorname*{Res}_{z=0} f(z) = \operatorname*{Res}_{z=0} \left(z^2 \sin\frac{1}{z}\right)$

$$z^2 \sin\frac{1}{z} = z^2 \left[\frac{1}{z} - \frac{1}{\lfloor 3} z^3 + \frac{1}{\lfloor 5} z^5 - \cdots \right]$$

$$= \left[z - \frac{1}{\lfloor 3} z + \frac{1}{\lfloor 5 z^2} - \cdots \right]$$

i.e. $b_1 = -\frac{1}{\lfloor 3} = -\frac{1}{6}$

Hence $\int_{|z|=1} z^2 \sin 1/z = 2\pi i \times \left(-\frac{1}{6}\right)$.

$= -\frac{\pi i}{3}$ #

(b) $\int_C \frac{5z^2}{(z-1)^2 (z+3)}$

Case (1) When $1, -3$ are inside C.

then $z=1 \rightarrow$ is a pole of order 2.

$z=-3 \rightarrow$ is a pole of order 1.

$\operatorname{Res}_{z=1} f(z) = \frac{\phi^{(1)}(1)}{\lfloor 1}$ where $\phi(z) = \frac{5z^2}{(z+3)}$

$\phi(1) \neq 0$

$\phi'(z) = \frac{10z(z+3) - 5z^2(1)}{(z+3)^2}$

$= \frac{10 \times 1 \times (4) - 5.1^2 . 1}{4^2}$

$= 35/16$

i.e. $\operatorname{Res}_{z=1} f(z) = \frac{35}{16}$

Now $\operatorname{Res}_{z=-3} f(z) = \phi(-3)$, where $\phi(z) = \frac{5z^2}{(z-1)^2}$

i.e. $\phi(-3) = \frac{45}{16}$

$= 45/16$.

Hence $\displaystyle\int_C \frac{5z^2}{(z-1)^2(z+3)}\,dz$

$$= 2\pi i \left(\operatorname*{Res}_{z=1} f(z)\right) + 2\pi i \operatorname*{Res}_{z=-3}\left(f(z)\right)$$

$$= 2\pi i \times \frac{35}{16} + 2\pi i \times \frac{45}{16}$$

$$= \frac{2\pi i}{16}\left[35+45\right]$$

$$= \frac{2\pi i}{\cancel{16}} \cdot \cancel{80}^{5}$$

$$= 10\pi i$$

Case ② When $1, -3$ are outside "C".

Then $\displaystyle\int_C \frac{5z^2}{(z-1)^2(z+3)}\,dz = 0$

by cauchy theorem.

(iii) $\displaystyle\int_{|z|=4} \frac{e^z}{(z+1)^{50}}\, dz$

i.e. $z=-1$ is a pole of order 50

$$\operatorname*{Res}_{z=-1} f(z) = \frac{\phi^{(49)}(-1)}{\lfloor 49} \quad \text{where } \phi(z) = e^z$$

$$= \frac{e^{-1}}{\lfloor 49} = \frac{1}{\lfloor 49\, e}$$

& hence $\displaystyle\int_{|z|=4} \frac{e^z}{(z+1)^{50}}\, dz = 2\pi i \times \frac{1}{\lfloor 49 \times e}$

$$= \frac{2\pi i}{\lfloor 49\, e}$$

(iv) $\displaystyle\int_{|z|=4} \frac{3z^3+2}{(z-1)(z^2+9)}\, dz$

$$= \int_{|z|=4} \frac{3z^3+2}{(z-1)(z+3i)(z-3i)}\, dz$$

i.e. $f(z) = \dfrac{3z^3+2}{(z-1)(z+3i)(z-3i)}$ has simple poles.

at $z=1$, $z=3i$, $z=-3i$.

$$\int_{|z|=4} \frac{3z^3+2}{(z-1)(z^2+9)}\, dz = 2\pi i \operatorname*{Res}_{z=1} f(z) + 2\pi i \operatorname*{Res}_{z=3i} f(z) + 2\pi i \operatorname*{Res}_{z=-3i} f(z)$$

Now

$$\operatorname*{Res}_{z=1} f(z) = \phi(1) \quad \text{where } \phi(z) = \frac{3z^3+2}{(z^2+9)}$$

$$= \frac{3\cdot 1^3+2}{1^2+9} = \frac{5}{10} = \underline{1/2}$$

$$\operatorname*{Res}_{z=3i} f(z) = \phi(3i) \quad \text{where } \phi(z) = \frac{3z^3+2}{(z-1)(z+3i)}$$

$$= \frac{3.(3i)^3+2}{(3i-1)(3i+3i)}$$

$$= \frac{-81i+2}{(3i-1)6i} \quad \#$$

$$\operatorname*{Res}_{z=-3i} f(z) = \phi(-3i), \quad \text{where } \phi(z) = \frac{3z^3+2}{(z-1)(z-3i)}$$

$$\phi(-3i) = \frac{3(-3i)^3+2}{(-3i-1)(-3i-3i)}$$

$$= \frac{81i+2}{(3i+1)6i} \quad \#$$

Hence soln is given by

$$= 2\pi i \times \frac{1}{2} + 2\pi i \times \frac{(-81i+2)}{(3i-1)(6i)} + 2\pi i \times \frac{(81i+2)}{(3i+1)6i}$$

#

(v) $$\int_{|z|=2} \frac{\cosh \pi z}{z(z^2+1)}\, dz$$

$$= 2\pi i \operatorname{Res}_{z=0} f(z) + 2\pi i \operatorname{Res}_{z=i} f(z) + 2\pi i \operatorname{Res}_{z=-i} f(z)$$

Now

$$\operatorname{Res}_{z=0} \left(\frac{\cosh \pi z}{z(z^2+1)} \right) = \left(\frac{\cosh \pi z}{z^2+1} \right)_{z=0} = 1$$

$$\operatorname{Res}_{z=i} \left(\frac{\cosh \pi z}{z(z^2+1)} \right) = \left(\frac{\cosh \pi z}{z(z+i)} \right)_{z=i} = 1/2$$

$$\operatorname{Res}_{z=-i} \left(\frac{\cosh \pi z}{z(z^2+1)} \right) = \left(\frac{\cosh \pi z}{z(z-i)} \right)_{z=-i} = \frac{1}{2}$$

$$\left(\text{use } \cosh z = \frac{e^z + e^{-z}}{2} \right)$$

Hence

$$\int_{|z|=2} \frac{\cosh \pi z}{z(z^2+1)} = 2\pi i \times 1 + 2\pi i \times \frac{1}{2} + 2\pi i \times \frac{1}{2}$$

$$= 2\pi i + \pi i + \pi i$$

$$= 4\pi i \quad \#$$